

Homework 4 — Sports Analytics (Tripti Singh)

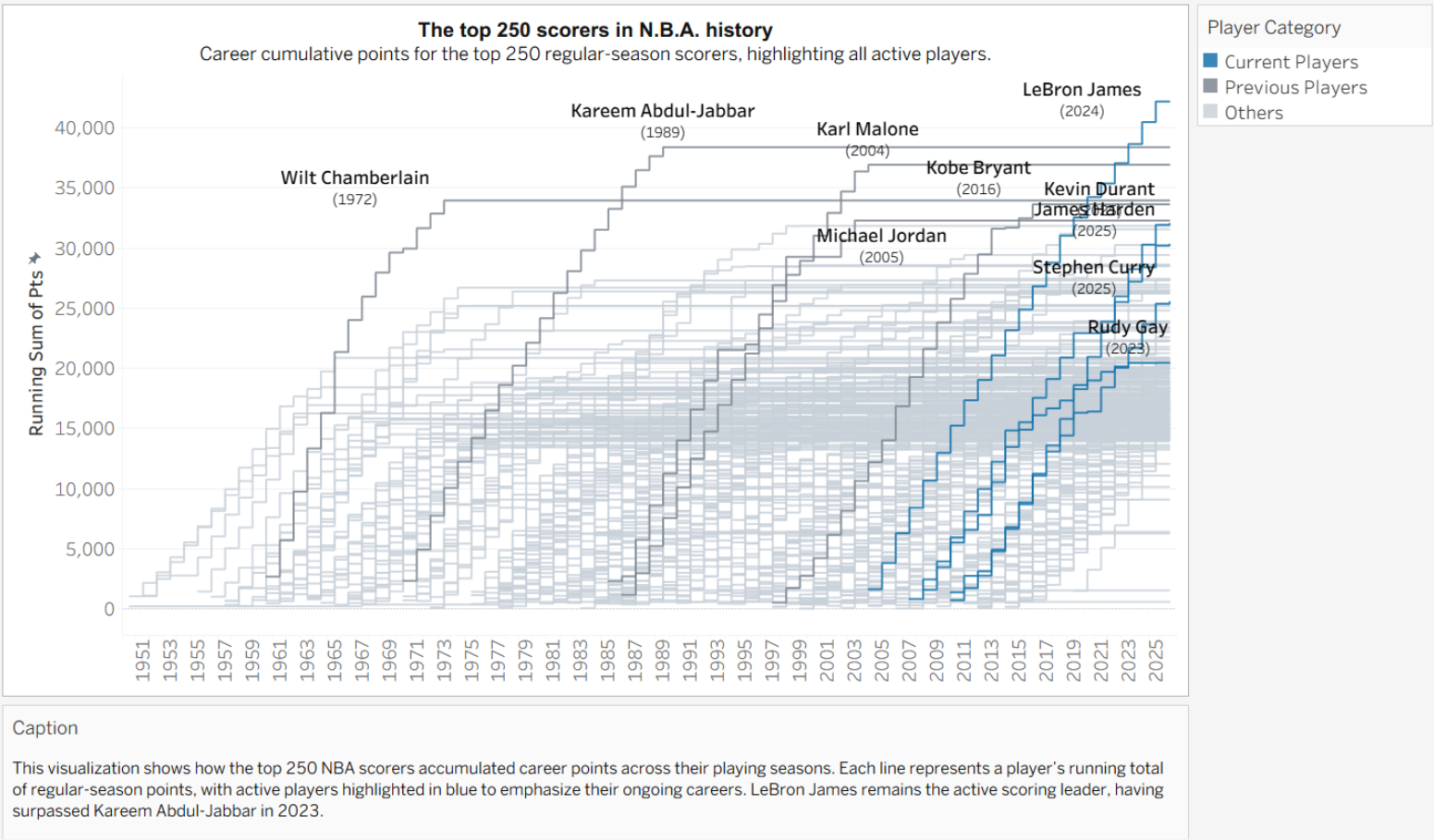
Part A — Recreate 3 of 6 NYT visuals (Tableau)

Data dictionary for the datasets – Player Totals

Column Name	Description	Type
season	The NBA season in which the statistics were recorded. Usually written as a single year (e.g., 2021) representing the season ending in that year.	Text
lg	Indicates the league the player competed in during that season: NBA (current), ABA (American Basketball Association), or BAA (pre-NBA). Useful for comparing players across eras.	Text
player	Full name of the player exactly as listed in official league records.	Text
player_id	A unique numerical/text identifier that distinguishes each player across seasons and teams. Useful for grouping and merging.	Text/Integer
age	The player's age during the listed season. Helps analyze career progression and age-related performance.	Integer
team	The abbreviation of the team the player was on for that season (e.g., LAL = Lakers, GSW = Warriors).	Text
pos	The primary playing position of the player for that season. Common values include PG (Point Guard) , SG (Shooting Guard) , SF (Small Forward) , PF (Power Forward) , and C (Center) . Positions help identify a player's role and style of play (e.g., guards shoot more 3s, centers score more 2s).	Text
g	Total number of games the player appeared in during the season.	Integer
gs	Total number of games the player started, indicating role importance and minutes stability.	Integer
mp	Total minutes played during the season (or per-game minutes in some versions).	Numeric
fg	Total field goals made (2-pointers + 3-pointers). Measures scoring volume.	Numeric
fga	Total field goals attempted. Useful for calculating efficiency and usage.	Numeric
fg_percent	Field goal percentage: $FG \div FGA$. Indicates overall shooting efficiency.	Percentage
x3p	Number of 3-point field goals made. Reflects perimeter scoring ability.	Numeric
x3pa	Number of 3-point field goals attempted. Indicates volume of outside shots taken.	Numeric
x3p_percent	3-point percentage: $3P \div 3PA$. Shows long-range shooting efficiency.	Percentage
x2p	Total 2-point field goals made. Important for evaluating interior or mid-range scoring.	Numeric
x2pa	Total 2-point field goals attempted. Helps calculate inside scoring efficiency.	Numeric
x2p_percent	2-point field-goal percentage: $2P \div 2PA$. Measures efficiency at the rim or mid-range.	Percentage
e_fg_percent	Effective Field Goal Percentage: adjusts FG% by giving extra weight to 3-pointers since they are worth more. Formula: $(FG + 0.5 \times 3P) \div FGA$.	Percentage
ft	Number of free throws made. An important component of scoring efficiency.	Numeric
fta	Number of free throws attempted. Helps measure aggressiveness and foul-drawing ability.	Numeric
ft_percent	Free-throw percentage: $FT \div FTA$. Shows accuracy from the free-throw line.	Percentage
orb	Offensive rebounds collected. Shows ability to extend possessions and create second-chance points.	Numeric
drb	Defensive rebounds collected. Indicates contribution to ending opponent possessions.	Numeric
trb	Total rebounds: offensive + defensive. Measures overall rebounding production.	Numeric
ast	Total assists recorded. Indicates passing ability and playmaking involvement.	Numeric
stl	Total steals made. Reflects defensive anticipation and on-ball pressure.	Numeric
blk	Total blocks made. Indicates rim protection and defensive impact near the basket.	Numeric
tov	Turnovers committed. Helps evaluate ball security and decision-making.	Numeric
pf	Personal fouls committed. Useful for assessing defensive discipline and playing time limitations.	Numeric
pts	Total points scored in the season (all FGs + FTs). Primary scoring metric.	Numeric
trp_dbl	Number of triple-doubles recorded in the season (10+ in 3 stat categories). Indicates all-around performance.	Integer

I handled the TOT rows by using the combined season totals instead of team-level splits, which prevents double-counting player statistics. I also ensured that missing 3-point values from older eras were treated as zero so that calculated shot-share fields (Share 2P and Share 3P) remain accurate for all players.

1. The top 250 scorers in NBA history. (visualize Career Cumulative Points) — LeBron vs Kareem (+ optional comparators).



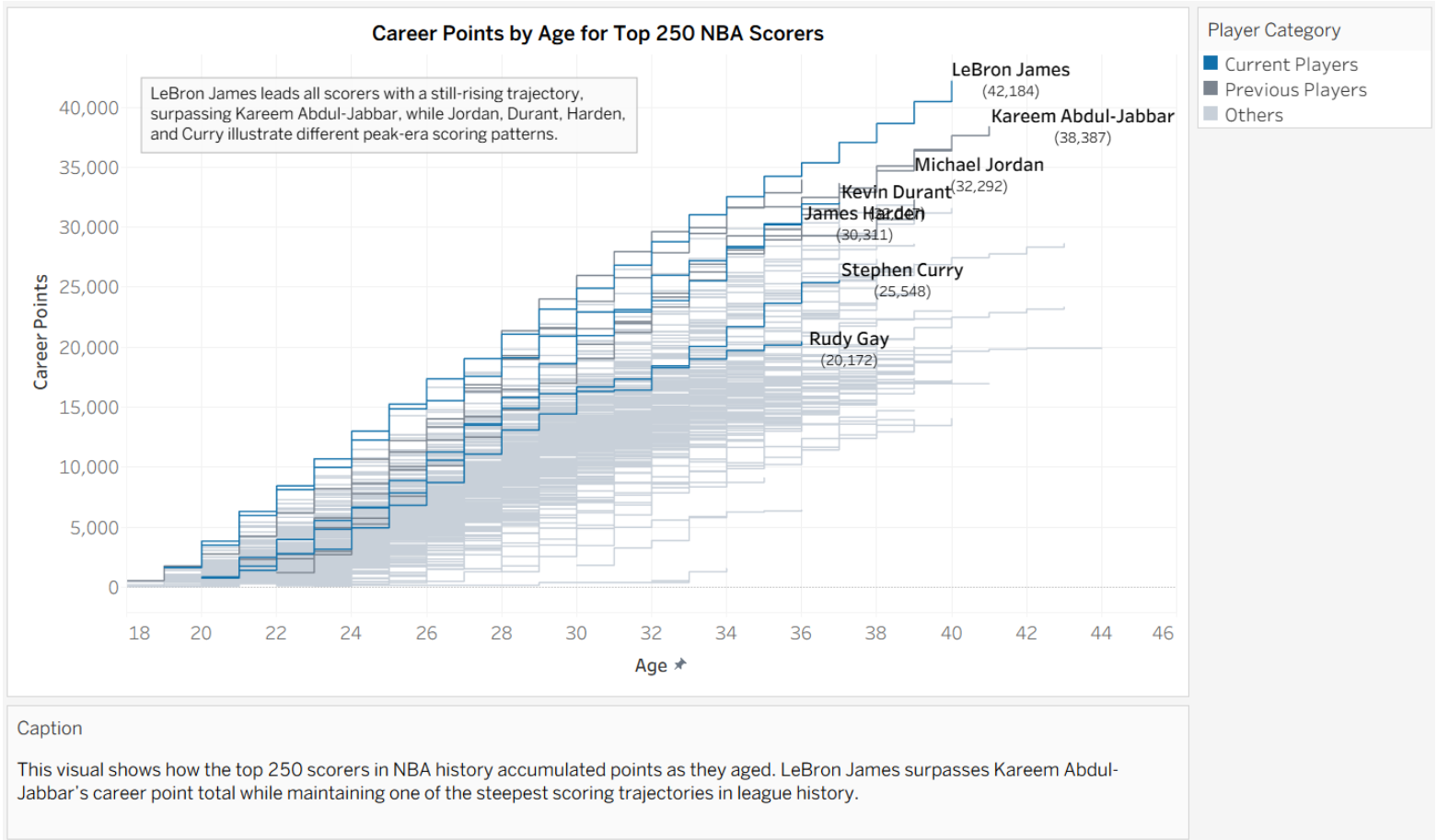
Methodology

- Filtered the dataset to include only **NBA** league seasons.
- Computed **career cumulative points** for each player using the *Running Sum* table calculation on season totals.
- Used a Set in Tableau (“Top 250 Scorers”) to isolate only players with the highest career total points.
- Created a **Player Category Group** to segment players into:
 - Current Players, Previous Players & Others
- Highlighted active players in blue and displayed all others in greyscale.
- Added key player annotations (LeBron, Kareem, Jordan, Malone, Chamberlain) to match the style of the NYT visual.
- Applied formatting to improve readability, gridlines, label spacing, custom colors, and subtitles.

Insights & Interpretation

- **LeBron James** (2024) remains the NBA’s all-time leading scorer, with his career line still rising.
 - **Kareem Abdul-Jabbar** (1989) previously held the record for nearly four decades,
 - **Michael Jordan** (2005) reached one of the highest scoring peaks in league history.
 - **Wilt Chamberlain** (1972) shows the steepest early-career scoring surge, reflecting his historic dominance.
- Together, these trajectories highlight different scoring styles, early peaks, long-term consistency, and modern longevity.

2. Career points by age for top 250 scorers.



Methodology

- Filtered NBA players only using the Lg field.
- Identified the top 250 career scorers by computing Career Points = SUM(PTS) per player.
- Filtered the dataset to include only these top 250 players.
- Plotted Age on the X-axis and season points (PTS) on the Y-axis.
- Created a Running Sum table calculation to obtain:
- Cumulative Career Points = RUNNING_SUM(SUM(PTS))

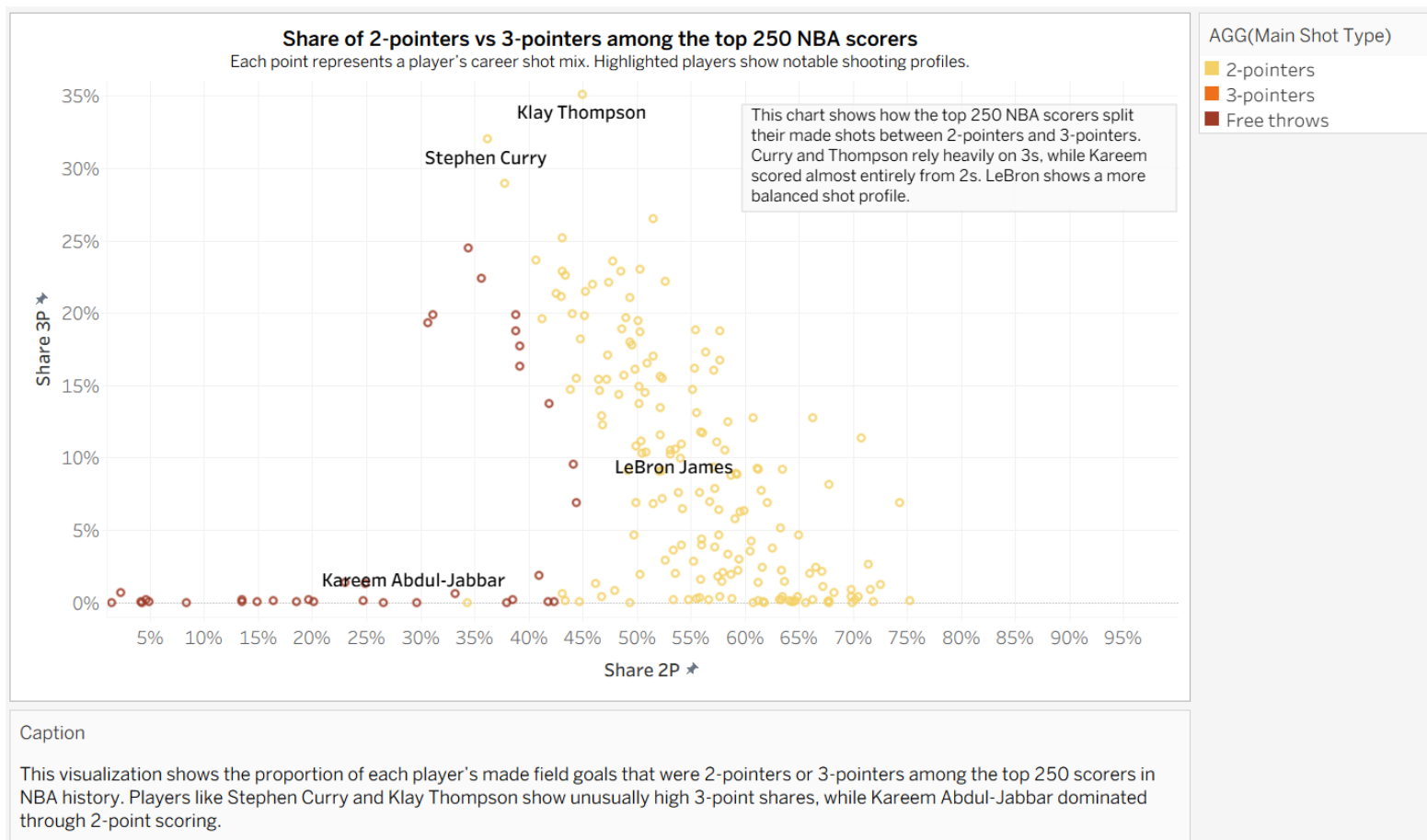
Computed by Age, restarting for every player.

1. Grouped players into three categories & Styled lines:
 - Current players, Previous all-time greats & Others
 - Blue for current players, Dark gray for legends & Light gray for others
2. Added labels for notable players (LeBron, Kareem, Durant, Curry, Harden, Rudy Gay).

Insights & Interpretation

- **LeBron James** has one of the steepest and longest career scoring curves, surpassing Kareem Abdul-Jabbar's historical record.
- **Kareem Abdul-Jabbar** maintained consistent scoring over 20 seasons, ending slightly below LeBron's final total.
- **Kevin Durant** and **James Harden** show strong mid-career peak scoring despite playing fewer seasons than LeBron or Kareem.
- **Stephen Curry** accumulates fewer total points because his career started slower and injuries reduced early-career totals.
- Many players plateau around ages 35–38, with very few scorings significantly past age 40.

3. Of the top 250 scorers, share of each player's shots that were..



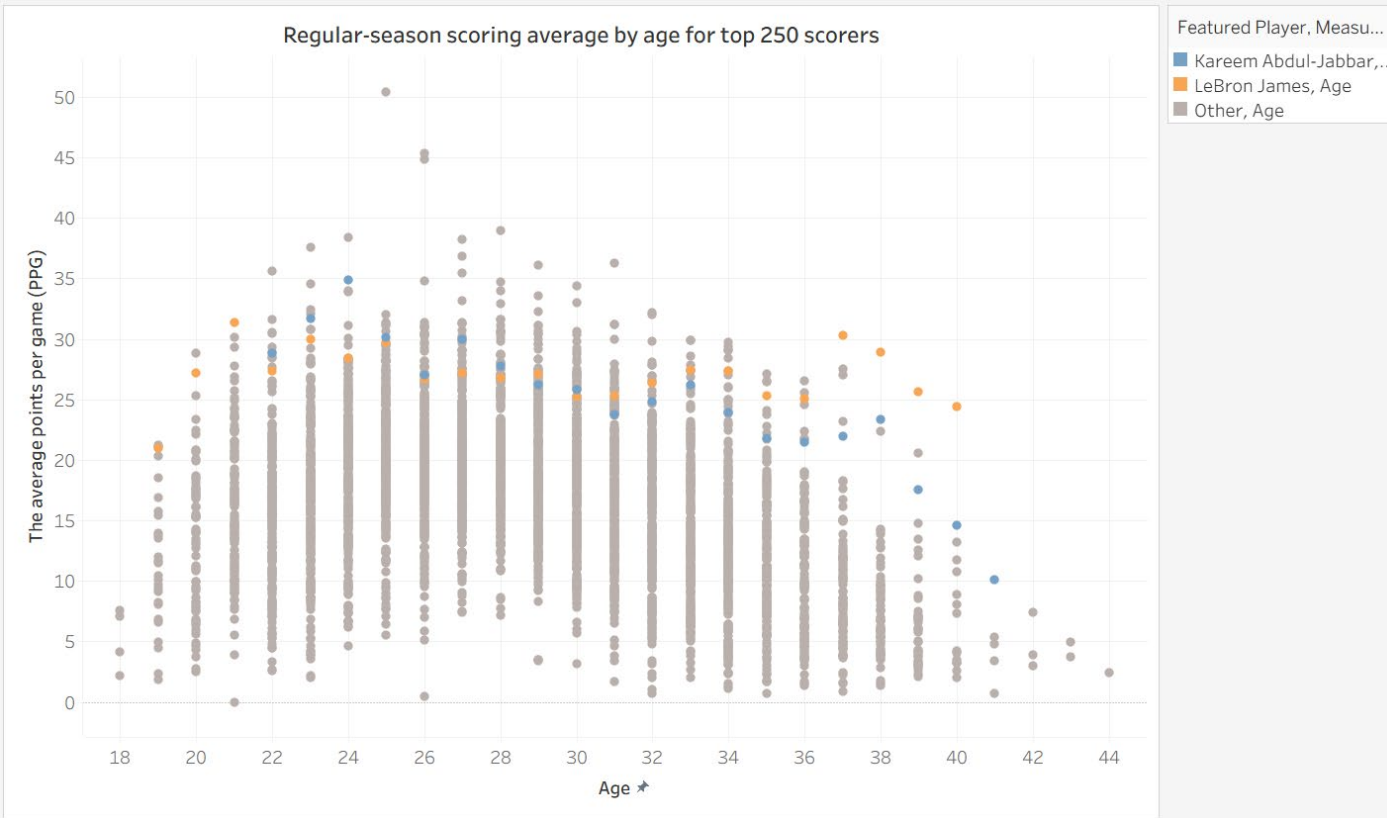
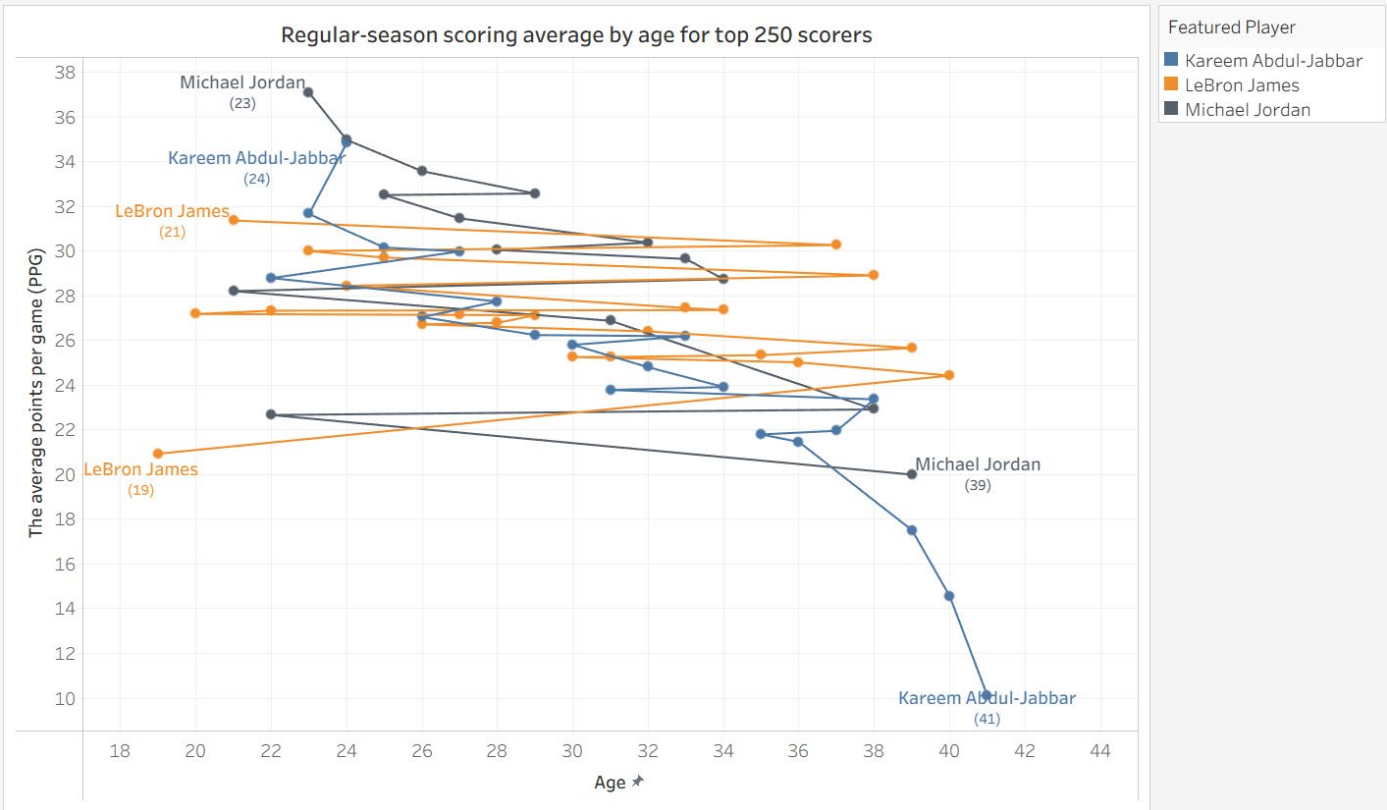
Methodology (Short Version)

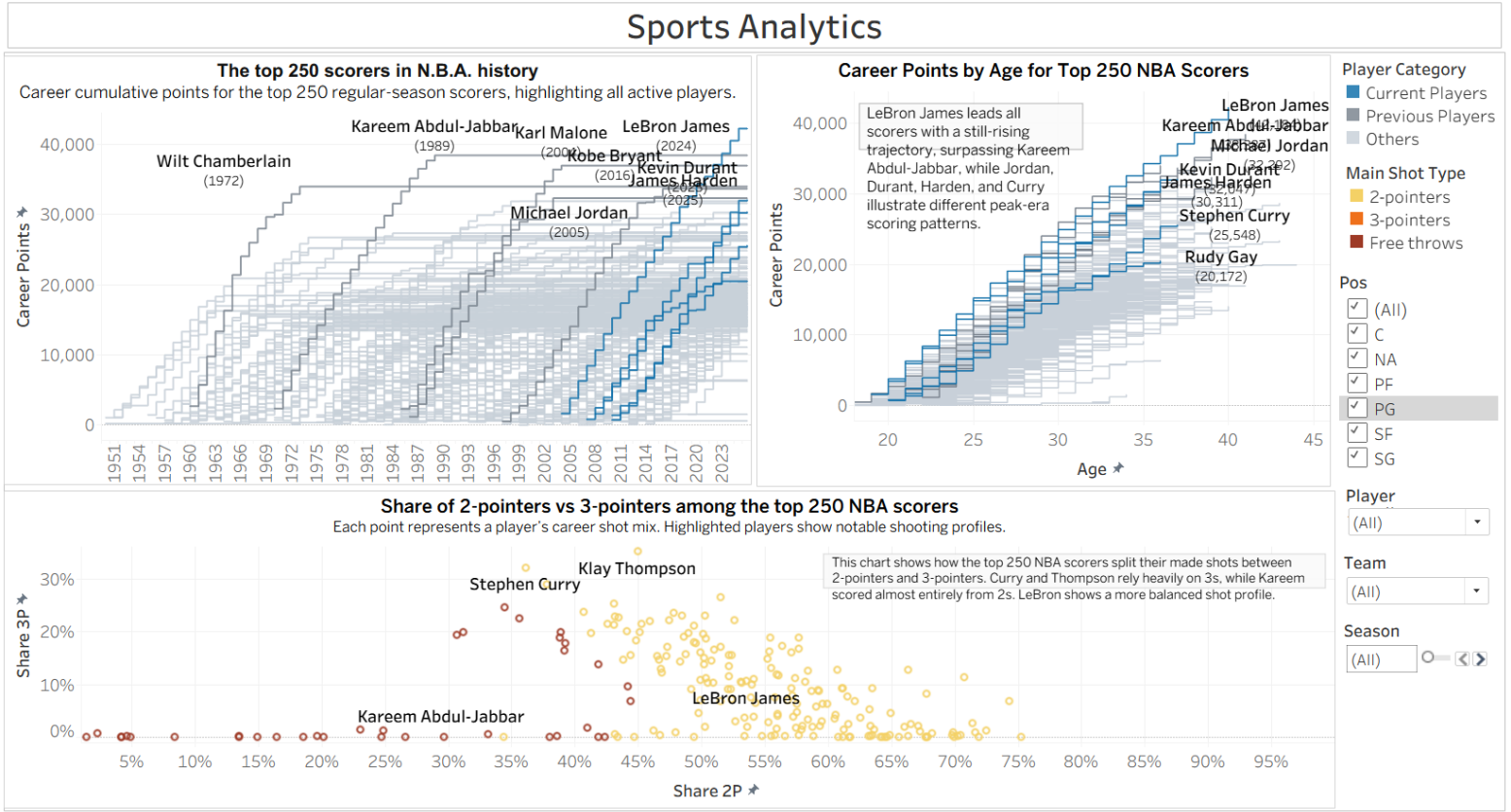
- To recreate this chart, I used the NBA player totals dataset and filtered the top 250 career scorers. For each player, I calculated their total 2-pointers, 3-pointers, and field goals. I then created two Tableau calculated fields:
 - Share 2P = $(2P \div FG)$
 - Share 3P = $(3P \div FG)$
- These percentages measure how much of each player's scoring came from 2-point vs 3-point shots.
- I aggregated values at the career level and built a scatterplot with Share 2P on the x-axis and Share 3P on the y-axis.
- Points were colored by each player's dominant shot type, and key players (Curry, Klay, LeBron, Kareem) were manually labeled for context.

Insights & Interpretation

- Most players in the top 250 scored mainly from 2-pointers, which is why so many dots are clustered on the right side of the chart.
- Stephen Curry** and **Klay Thompson** are clear outliers with very high 3-point shares, showing how much they rely on perimeter shooting.
- Kareem Abdul-Jabbar** is almost entirely a 2-point scorer, which makes sense because he played inside and never took 3s.
- LeBron James** sits in the middle, showing a balanced scoring profile with both 2-pointers and 3-pointers.
- Overall, the chart shows how scoring styles have changed over time, from inside scoring (**Kareem**) to modern 3-point heavy players (Curry & Klay).

4. Regular-season scoring average by age for top 250 scorers (Tried Creating 4 visual just to learn about it)



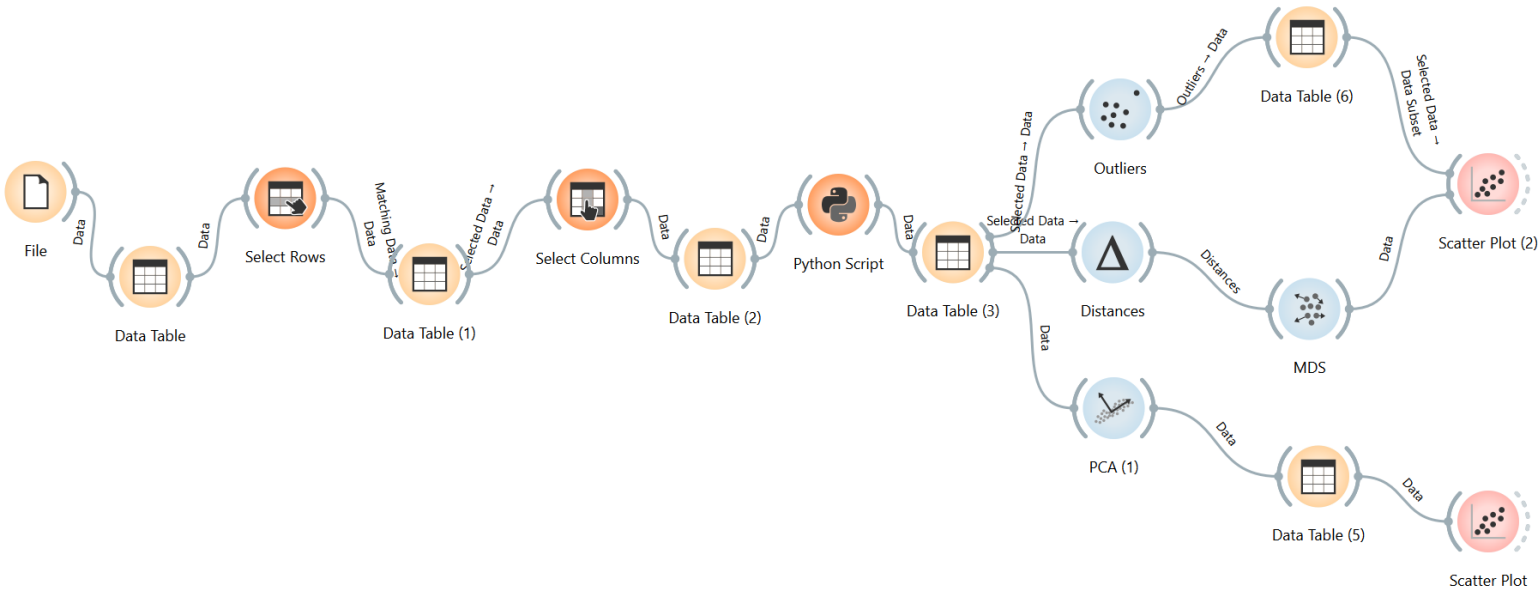


Part B — Similarity & Outliers

Data dictionary for the datasets – Per 36 Minutes

Column Name	Description	Type
season	The NBA season in which the statistics were recorded. Usually written as a single year (e.g., 2021) representing the season ending in that year.	Text
lg	Indicates the league the player competed in during that season: NBA (current), ABA (American Basketball Association), or BAA (pre-NBA). Useful for comparing players across eras.	Text
player	Full name of the player exactly as listed in official league records.	Text
player_id	A unique numerical/text identifier that distinguishes each player across seasons and teams. Useful for grouping and merging.	Text/Integer
age	The player's age during the listed season. Helps analyze career progression and age-related performance.	Integer
team	The abbreviation of the team the player was on for that season (e.g., LAL = Lakers, GSW = Warriors).	Text
pos	The primary playing position of the player for that season. Common values include PG (Point Guard) , SG (Shooting Guard) , SF (Small Forward) , PF (Power Forward) , and C (Center) . Positions help identify a player's role and style of play (e.g., guards shoot more 3s, centers score more 2s).	Text
g	Total number of games the player appeared in during the season.	Integer
gs	Total number of games the player started, indicating role importance and minutes stability.	Integer
mp	Total minutes played during the season (or per-game minutes in some versions).	Numeric
fg_per_36_min	Field goals made per 36 minutes. Measures scoring efficiency independent of playing time.	Numeric
fga_per_36_min	Field goals attempted per 36 minutes. Reflects shooting volume.	Numeric
fg_percent	Field-goal accuracy (FG%): $FG \div FGA$. Shows general shooting efficiency.	Percentage

Column Name	Description	Type
x3p_per_36_min	3-pointers made per 36 minutes. Indicates perimeter scoring output.	Numeric
x3pa_per_36_min	3-pointers attempted per 36 minutes. Measures outside shooting frequency.	Numeric
x3p_percent	3-point accuracy (3P%): $3P \div 3PA$.	Percentage
x2p_per_36_min	2-pointers made per 36 minutes. Reflects interior/mid-range scoring.	Numeric
x2pa_per_36_min	2-pointers attempted per 36 minutes.	Numeric
x2p_percent	2-point accuracy (2P%): $2P \div 2PA$.	Percentage
e_fg_percent	Effective FG%, adjusts FG% by giving extra credit for 3-pointers. Formula: $(FG + 0.5 \times 3P) / FGA$.	Percentage
ft_per_36_min	Free throws made per 36 minutes.	Numeric
fta_per_36_min	Free throws attempted per 36 minutes.	Numeric
ft_percent	Free-throw accuracy (FT%). Indicates reliability at the line.	Percentage
orb_per_36_min	Offensive rebounds per 36 minutes. Measures second-chance creation.	Numeric
drb_per_36_min	Defensive rebounds per 36 minutes. Shows ability to end opponent possessions.	Numeric
trb_per_36_min	Total rebounds (offensive + defensive) per 36. Overall rebounding productivity.	Numeric
ast_per_36_min	Assists per 36 minutes. Indicates playmaking impact.	Numeric
stl_per_36_min	Steals per 36 minutes. Reflects defensive pressure and anticipation.	Numeric
blk_per_36_min	Blocks per 36 minutes. Represents rim protection.	Numeric
tov_per_36_min	Turnovers per 36 minutes. Measures ball-handling risk and decision errors.	Numeric
pf_per_36_min	Personal fouls per 36 minutes. Shows defensive discipline level.	Numeric
pts_per_36_min	Points scored per 36 minutes. Core scoring output normalized for playing time.	Numeric



Python Script for Ranking Players by Importance Score (Top 100 Extraction)

- This script takes the dataset from Orange, converts it into a pandas DataFrame, removes unnecessary meta columns, and calculates a new Importance Score using per-36 stats.
- The score rewards scoring, playmaking, rebounding, and defense, while penalizing turnovers.
- After computing the score, the script selects the Top 100 most impactful players, rebuilds a new Orange dataset with all original stats plus the new importance score, and outputs it for further analysis.

```
import numpy as np
import pandas as pd
from Orange.data import Table, Domain, ContinuousVariable

# ----- STEP 1: Convert Orange table -> pandas -----
attr_names = [var.name for var in in_data.domain.attributes]
df_attrs = pd.DataFrame(in_data.X, columns=attr_names)
# Metas (player, team, selected, etc.)
meta_names = []
df metas = pd.DataFrame()
if in_data.domain.metas:
    meta_names = [m.name for m in in_data.domain.metas]
    df metas = pd.DataFrame(in_data.metas, columns=meta_names)
# Combine metas + attributes
df = pd.concat([df metas, df_attrs], axis=1)
# Lowercase everything
df.columns = [c.lower() for c in df.columns]
# ----- STEP 2: Remove any "selected" meta column -----
selected_cols = [c for c in df.columns if "selected" in c]
df = df.drop(columns=selected_cols, errors="ignore")
# Clean meta names accordingly
meta_names = [m for m in meta_names if m.lower() not in selected_cols]
# ----- STEP 3: Stat columns -----
points = "pts_per_36_min"
assists = "ast_per_36_min"
rebounds = "trb_per_36_min"
steals = "stl_per_36_min"
blocks = "blk_per_36_min"
turnovers = "tov_per_36_min"
# ----- STEP 4: Importance Score -----
df["importance_score"] = (
    0.40 * df[points] +
    0.20 * df[assists] +
    0.15 * df[rebounds] +
    0.15 * (df[steals] + df[blocks]) -
    0.10 * df[turnovers]
)
# ----- STEP 5: Select TOP 100 -----
df_top100 = df.sort_values("importance_score", ascending=False).head(100)
# ----- STEP 6: Build new Orange domain -----
new_attrs = list(in_data.domain.attributes) + [ContinuousVariable("importance_score")]
new_meta_vars = [m for m in in_data.domain.metas if m.name.lower() not in selected_cols]
new_domain = Domain(new_attrs, in_data.domain.class_vars, new_meta_vars)
# ----- STEP 7: Construct Orange Table -----
# Attributes
X = df_top100[[var.name for var in in_data.domain.attributes]].to_numpy()
# Add new column
importance_col = df_top100[["importance_score"]].to_numpy()
```



```
X = np.hstack([X, importance_col])
```

```
# Targets
```

```
Y = in_data.Y[:100] if in_data.Y is not None and len(in_data.Y) > 0 else None
```

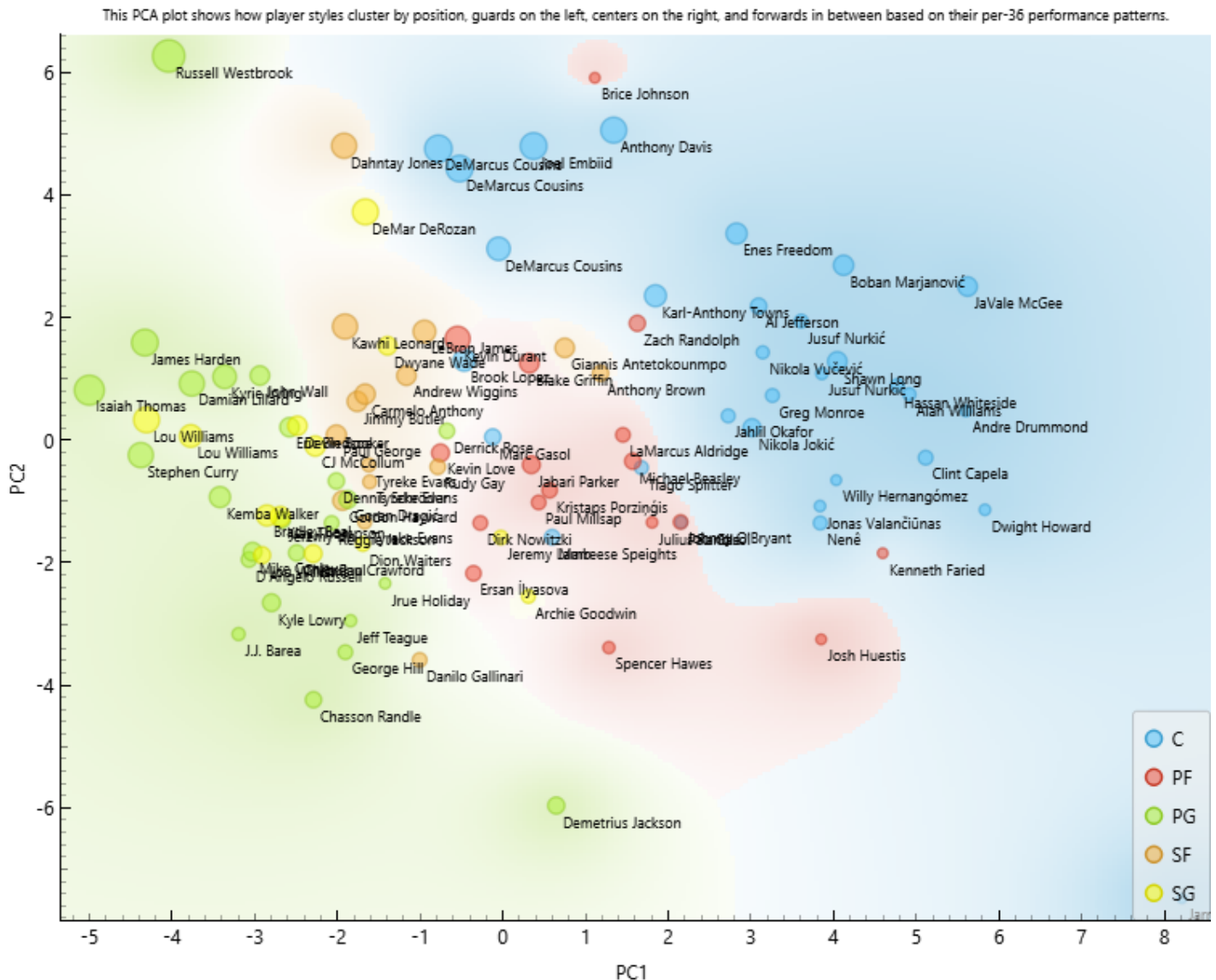
```
# Metas
```

```
M = df_top100[[m.name.lower() for m in new_meta_vars]].to_numpy() if new_meta_vars else None
```

```
# ----- STEP 8: Output -----
```

```
out_data = Table(new_domain, X, Y, M)
```

1. Recreate PCA1 vs PCA2 scatter

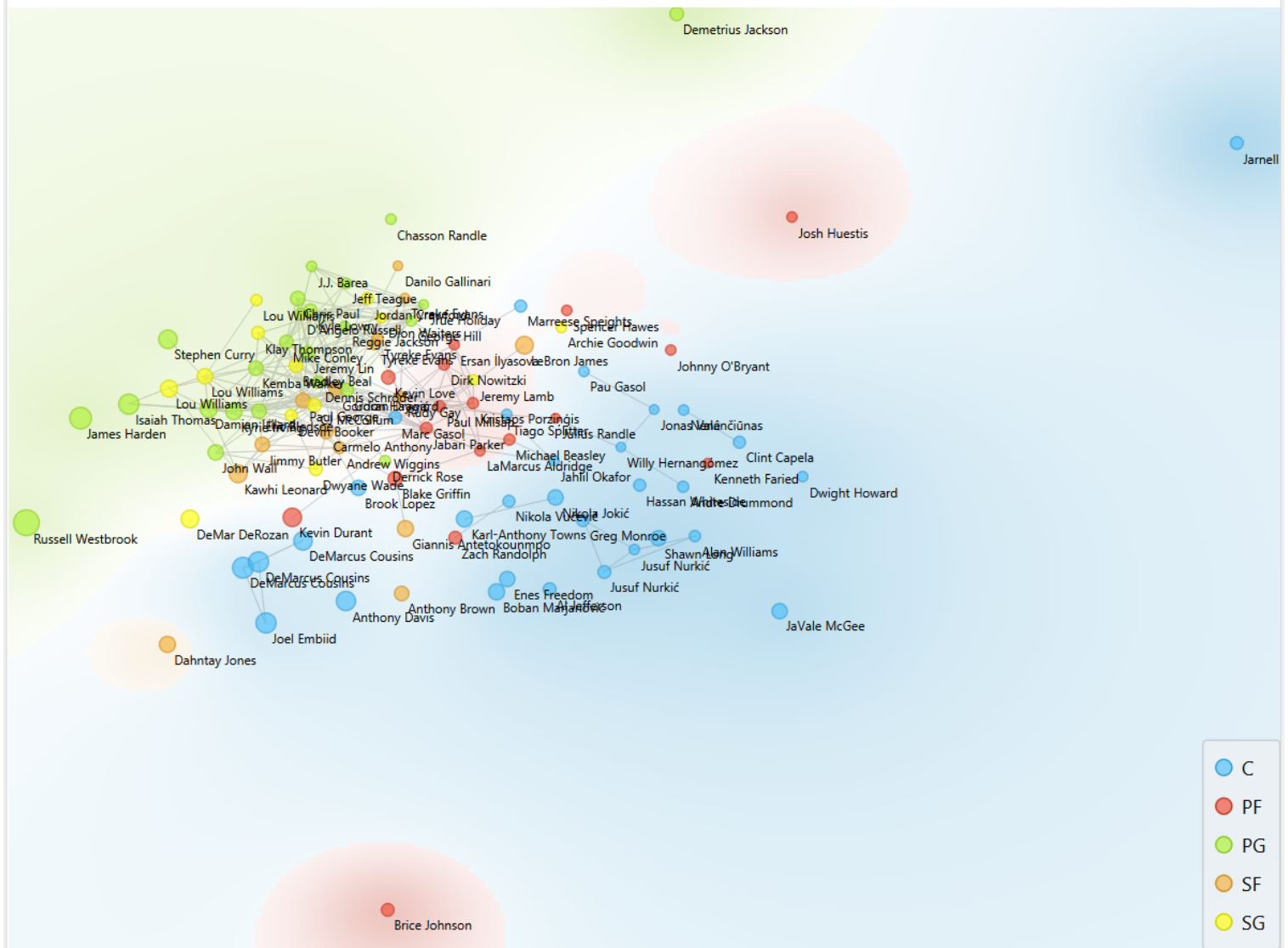


Interpretation

- The PCA naturally separates player styles: guards (like **Stephen Curry**, **James Harden**) cluster on the left due to high assists, steals, and perimeter scoring, while centers (like **Andre Drummond**, **Rudy Gobert**) appear on the right because of rebounding and shot-blocking stats.
- Versatile forwards (such as **LeBron James**, **Kevin Durant**) fall near the center, reflecting balanced inside-outside production.
- Outliers like **Russell Westbrook** (high usage + assists) and **Joel Embiid** (elite scoring + defense) sit farther from their groups, showing unique statistical profiles.
- Overall, the PCA proves that player positions and styles emerge clearly from per-36 data, even without using position labels.

2. MDS projection

MDS Map of Player Similarities by Position (Per-36 Performance)



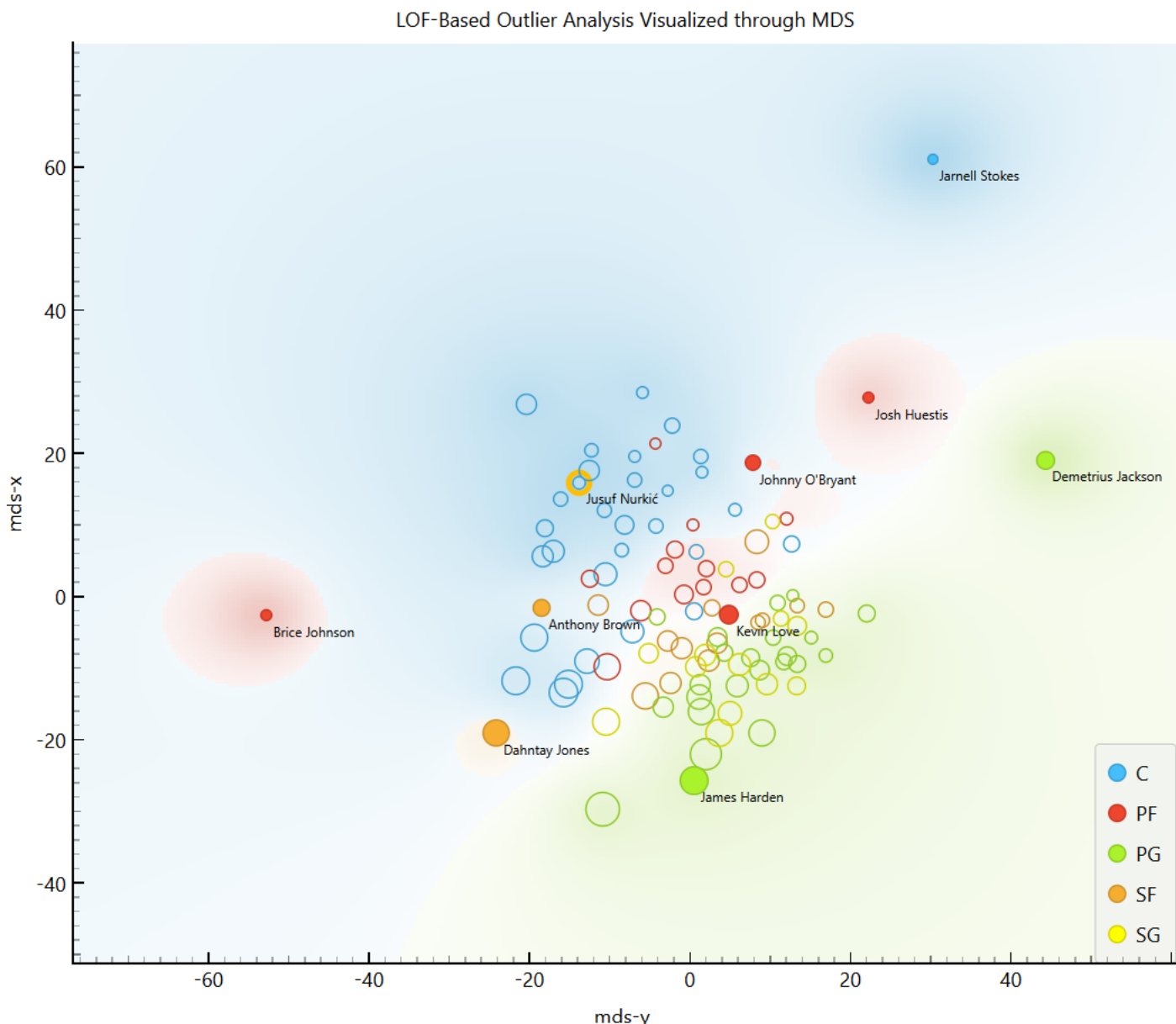
Interpretation:

- The MDS map shows how similar NBA players are based on their performance per-36. Most players gather tightly in the center, meaning their overall stat profiles are similar.
- Guards like **Stephen Curry, James Harden, Lou Williams, and Isaiah Thomas** cluster lower on the map, reflecting guard-style scoring and playmaking.
- Big men such as **Dwight Howard, Javale McGee, Hassan Whiteside, and Enes Freedom** group higher, showing strong rebounding and interior production.
- A few players appear far from everyone else **Jarnell Stokes** at the top, **Josh Huestis** on the right, and **Brice Johnson** on the far left indicating unique or uncommon statistical patterns compared to the league.
- Overall, MDS shows natural position-based grouping, while also highlighting rare outliers with distinct play styles.

Comparison: How Clusters Differ Between PCA and MDS –

- PCA creates a much clearer separation of player types guards like **Curry, Lillard, Harden** gather on one side, bigs like **Towns, Embiid, Jokic** spread to the opposite side, and hybrid forwards such as **LeBron and Durant** fall in the middle.
- PCA stretches the data along directions of maximum variance, so extreme scorers or size-driven players appear farther apart.
- MDS, on the other hand, forms a tighter, more rounded cloud. Most players cluster near the center, and only unusual profiles like **Jarnell Stokes** or **Brice Johnson** become isolated. Bigs still group together and guards stay relatively close, but the boundaries between positions are softer.
- Overall: **PCA exaggerates positional differences, while MDS focuses on overall similarity**, causing players to appear more blended with only the most unique ones standing out.

3. Outlier analysis via Local Outlier Factor (LOF)



Interpretation:

Using LOF with a 10% contamination setting, Orange identifies roughly the top 10% of players whose per-36 stats are statistically unusual compared to the rest of the dataset. In the MDS map, these outliers appear far away from the dense central cluster, showing players whose scoring, usage, minutes, or efficiency patterns differ sharply from typical positional profiles.

Top 5 Outliers Identified:

1. **Jarnell Stokes** – extreme outlier placed at the very top; his per-36 production is unusually efficient relative to extremely limited minutes.
2. **Brice Johnson** – isolated far left; very low usage and irregular stat distribution make him statistically inconsistent with PF averages.
3. **Josh Huestis** – high separation on the right; defensive-heavy profile with minimal scoring creates a rare signature.
4. **Demetrius Jackson** – distant green cluster; extremely limited playing time with sharp variance in assists/turnovers.
5. **Dahntay Jones** – positioned far below the main group; old-career, low-minutes profile causes instability in per-36 scaling.

Overall, the LOF analysis shows players whose unbalanced or minutes-inflated per-36 numbers place them far from the league's normal patterns. The MDS layout helps visualize this clearly, with outliers forming small, isolated pockets away from the main cloud of guards, forwards, and centers.